Benton Tripp

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Experienced data scientist and geospatial analyst with a strong background in data analytics, machine learning, statistical modeling, geospatial science, and applied software development. I bring over six years of professional experience across local government, utilities, consulting, marketing analytics, and forecasting, with a focus on building analytical tools, models, dashboards, and data workflows that support operational and strategic decisions. Known for self-motivation, clear communication, and the ability to learn and apply new methods efficiently, I deliver practical, well-documented, and defensible analytical products in both team-oriented and independent settings.

Work Experience

Highly skilled in technical problem-solving and data analysis, I leverage my expertise in machine learning, statistical modeling, and geospatial technologies to support business objectives. My technical competencies span advanced R and Python programming, cloud computing (AWS, Azure), and deployment of machine learning pipelines using MLOps platforms. As a natural leader and team player, I consistently deliver high-quality results, providing valuable insights and fostering a collaborative environment to drive success.

Technology Analyst

City of Raleigh
Raleigh, NC
June 2025 - Present

- Evaluated department data systems and analytical processes to identify improvement opportunities and implement data-driven solutions supporting Planning and Development operations.
- Built a PyTorch-based computer vision application that automatically detects and redacts engineering seals and other sensitive information in large PDF plan sets and permit documents, reducing manual review effort.
- Produced city population estimates and long-range forecasts using statistical and spatial modeling methods that integrate census data, housing inventories, and permit activity.
- Designed and developed public-facing Power BI dashboards that allow residents and developers to track review timelines, benchmark compliance, and monitor development review performance.

- Developed interactive dashboards providing citywide performance metrics across planning, annexations, permitting, inspections, and related development processes.
- Created and maintained more than 20 internal Power BI dashboards supporting operational reporting, performance monitoring, workload management, and decision-making across multiple divisions.
- Performed analyses of development and demographic trends, including citywide suburban and urban growth analysis presented to City Council and senior leadership.
- Developed automated workflows that streamlined department operations, including meeting tracking, event management, and customer intake processing.
- Prepared technical reports and documentation, and provided training and support to staff on dashboards, data interpretation, and visualization best practices.

Data Scientist

Tennessee Valley Authority
Chattanooga, TN (Remote)
August 2022 - June 2025

- Developed end-to-end machine learning and analytical solutions across environmental, geochemical, and operational domains, including data pipelines, statistical models, and interactive R Shiny applications.
- Built large-scale analytical applications for groundwater evaluation, geochemical modeling, and billing anomaly detection, improving access to complex analytical workflows for business users.
- Developed document analysis tools using large language models, retrieval-augmented generation (RAG), embeddings, and LangChain to support search, extraction, and review of internal documents.
- Led technical proof-of-concepts using Python, R, SQL, AWS, and version control to evaluate new analytics use cases and translate business problems into working prototypes.
- Participated in an enterprise MLOps platform assessment covering Domino Data Lab, DataRobot, Dataiku, SAS, and AWS SageMaker, with a focus on model monitoring, reproducibility, and workflow management.
- Used Azure DevOps, AWS services, and Dataiku to develop, deploy, and maintain scalable analytical and machine learning workflows.
- Developed and instructed advanced R programming courses, helping technical and non-technical stakeholders better understand analytical methods and tools.

Data Analyst

Elder Research Inc.
Raleigh, NC
August 2021 - August 2022

- Built a machine learning pipeline using Databricks for ETL, model training, and deployment that identified major slippage drivers for Church & Dwight, resulting in over $1 million in savings.
- Designed and implemented a production-grade data engineering workflow for Chick-fil-A using AWS Redshift, Athena, and Apache Airflow, automating reconciliation between inventory and sales forecasting systems.
- Developed predictive analytics solutions for commercial clients, including classification, forecasting, recommendation, and data integration workflows.
- Used Python, R, SQL, cloud data platforms, version control, and reproducible development practices to build reliable analytical datasets and model workflows.
- Documented methods, maintained code quality, and communicated findings to technical and business teams.

Data Science Intern

Merkle Inc.
New York City, NY (Remote)
June 2021 - August 2021

- Developed customer segmentation and propensity modeling workflows for targeted marketing campaigns using Python, Databricks, and statistical modeling methods.
- Built and optimized data pipelines for the 2021 Jaguar/Land Rover campaigns, including audience overlap analysis that improved targeting efficiency by 15%.
- Prepared features, trained models, generated propensity scores, and supported campaign analytics for data-driven audience selection.
- Used reproducible workflows in a production-oriented analytics environment to support model versioning, collaboration, and deployment readiness.

Forecasting Analyst

Genworth
Lynchburg, VA
January 2020 - March 2021

- Analyzed call center data and developed time-series forecasting models to predict future workload and staffing requirements.
- Used Python, SQL, and PostgreSQL to model incoming and outgoing call volume, capturing trends, seasonality, day-of-week patterns, and other factors affecting workload.
- Supported quantitative research and forecasting updates in response to the COVID-19 pandemic.
- Prepared analysis and communicated results to support staffing, scheduling, and operational planning decisions.

Education

Bachelor of Science in Data Analytics

- Western Governor's University
- Graduation: Fall, 2021

Graduate Certificate in Statistics

- North Carolina State University
- Completion: Spring, 2024

Master of Science in Geospatial Information Science and Technology

- North Carolina State University
- Graduation: Spring, 2025

Skills

Programming Languages

- Python
- R
- SAS
- SQL/NoSQL
- JavaScript/HTML/CSS

Software/Technologies

- Git
- Unix/Linux
- AWS (Athena, Redshift, EC2, Lambda)
- Dashboards (Power BI, Shiny, Dash)
- GIS Software (Esri Products, QGIS, GRASS GIS)
- MLOps Software (Dataiku, DataRobot, MLflow, Domino Data Lab)

Other Skills/Experience

- ETL and Data Integration
- Statistics & Applied Mathematics
- Machine Learning
- Deep Learning (PyTorch, TensorFlow, Keras)
- Natural Language Processing (NLP)
- LLMs (Retrieval-Augmented Generation, LangChain)
- Forecasting
- Geospatial Analytics & Spatial Statistics

Certifications/Achievements

- AWS Cloud Practitioner, July 2024 - *Amazon Web Services (AWS)*
- IT Information Library Foundations Certification (ITIL), June 2021 - *AXELOS Global Best Practice*

Independent Research/Projects

Species Distribution Modeling

- Conducted comparative analysis between machine learning and traditional approaches (Inhomogeneous Poisson Point Process, MaxEnt) for Species Distribution Modeling, implementing all analytical pipelines in R.
- Analyzed distribution patterns of 8 bird species across 4 US states using bioclimatic variables, developing fully reproducible workflows with robust version control.

Utilizing the Clay Foundation Model and Sentinel-2 Imagery for Urban Growth Monitoring

- Developed proof-of-concept study integrating Sentinel-2 multispectral satellite imagery with the Clay Foundation Model (open-source deep Earth Observation foundation model) to monitor urban growth patterns using Python and PyTorch.
- Engineered a scalable, data-efficient framework generating urban imperviousness estimates with higher temporal resolution than standard datasets like the National Land Cover Database.
- Processed Sentinel-2 imagery into structured geospatial datasets and leveraged Clay Foundation Model for feature extraction and embedding generation.
- Designed and evaluated both simple and complex neural network architectures in PyTorch, enhancing model performance through systematic hyperparameter optimization to achieve high accuracy in urban imperviousness prediction.
- Demonstrated methodology's potential for enhancing urban monitoring frequency and responsiveness, directly supporting sustainable urban planning and resource management initiatives.

Personal Attributes

With a strong work ethic and a results-driven mindset, I am committed to personal and professional growth. I thrive in dynamic environments that challenge me to push the boundaries of my knowledge and skill set. Highly organized and detail-oriented, I am passionate about continuously learning and applying new technologies and analytical approaches to solve complex problems and contribute to organizational success.